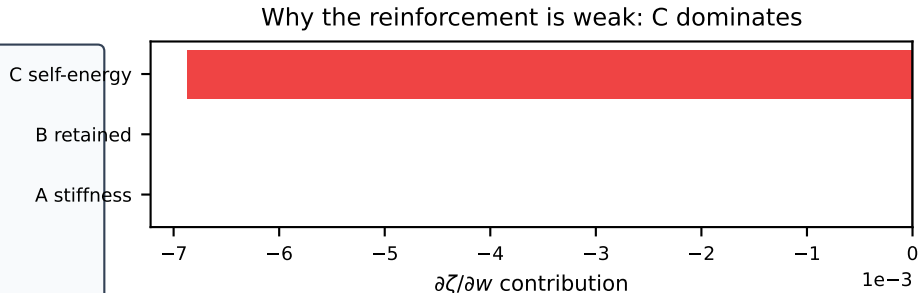


Weak-link report
Case: Mix60/no-PSS
Action: strengthen Line 20-34 by 1%

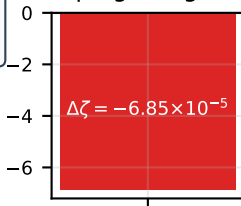
Classical view:
graph / stiffness terms nearly silent
 $A = -1.09\text{e-}9$, $B = +7.97\text{e-}9$

Controller-aware view:
PLL-dominated control pole is close
self-energy term $C = -6.87\text{e-}3$
control fraction = 0.959

Decision:
do not reinforce this line alone;
retune/control-separate PLL pole or add damping.



Damping margin falls



Control pole gets closer

